

An Automated Skill Extraction and Workload-Based Resource Recommendation System for WEA

A Proposed Capstone Project
presented to the College of Computer
Studies of Pamantasan ng Lungsod ng
Pasig

In partial fulfillment of the requirements for the degree of Bachelor of
Science in Information Technology

by:

Alpuerto, Lynn Czyla M.

Ebuen, Romell J.

Soriano, Vincent Miguel P.

Rodolfo A. Mirabel Jr., MT
Adviser

Implementation: AY 2026-2027

TABLE OF CONTENTS

CHAPTER 1: Introduction

- 1.1 The Problem and its Background
- 1.2 Review of Related Literature
- 1.3 Significance of the Study
- 1.4 Statement of the Problem
- 1.5 Scope and Limitation of the Study

CHAPTER 2: Methodology

- 2.1 Introduction
- 2.2 Research Design
 - Architectural Framework
 - Level 0 DFD/Schematic Diagram
 - Algorithm Discussion
 - SE Paradigm
- 2.3 Population and Sampling
- 2.4 Data Collection Methods
- 2.5 Ethical Considerations

CHAPTER 1

INTRODUCTION

The Problem and its Background

An effective resource management is one of the important for the success of any organization mostly in project based industries. It involves assigning the right employee to the right task that will be based on their skills and qualifications. The process is still done manually by reviewing resumes and certifications in many companies. But this method is slow, inconsistent and prone to human error. Organizations grow and will handle more projects, manual resource allocation becomes less efficient and harder to manage.

The technology becomes more essential in improving decision making processes in the increasing amount of data in organizations. According to Roy and Dutta (2022), a recommender system can help the organization to make better decisions by providing accurate suggestions in analyzing data. By adding information, Wang et al. (2023) discuss that using advanced recommendation can improve accuracy of matching people to a specific task or requirements. In the Philippines context, Cacatian et al. (2025) point out that using automation is becoming necessary for companies in Metro Manila to become more competitive and productive.

This study focuses on the WEA, a company that operates in the industrial distribution industry that serves clients in hazardous and highly regulated

environments, including oil refineries, petrochemical plants and other industrial facilities. The company primarily distributes specialized electrical products and equipment designed for hazardous area applications where safety and compliance are critical. The product distribution in WEA provides engineering services such as electrical system design, technical drawings, engineering calculations and project implementation support. Because these projects involve complex technical requirements and strict industry standards, employees must possess specific competencies, certifications and relevant experience before they can be assigned to project activities. .With the highly specialized nature of these projects, assigning employees to without the appropriate qualifications can lead not only to operational inefficiencies but also to significant safety risks. Therefore, there is a need for a system that can recommend qualified and available employees based on the project requirements.

The proposed An Automated Skill-Extraction and Workload-Based Resource Recommendation System is designed to improve workforce allocation at WEA by providing recommendations based on the employee qualifications, experience, certification, workload and availability. The system assists Resource Managers in identifying the most suitable personnel for project assignment while it promotes balanced resource utilization across the organization. To support these recommendations, the system utilizes OCR and NLP technologies to automate the extraction and processing of the employee skills and certification data from internal documents. At the same time, project managers can instantly have a list of available employees who meet the strict technical requirements of

hazardous environments. It prevents operational delays and ensure that qualified personnel are deploy to high risk sites. For the employees, the system will guarantees fair and transparent selection process, assignment are based in verified skills and availability rather than manual selection. Finally, this technology shift is the key for WEA to maintain the high project quality and safety standards improving overall productivity in competitive industries market.

Review of Related Literature

Recommender Systems and Fairness in Resource Allocation

Instead of making the recommendations in just accurate, the modern technology has been shifted in making them also fair and unbiased. According to Jin et al. (2023) and Li et al. (2022), the good system must be designed to give equal chance to all the employees, which is the first need that the study identifies, the need for a fair and unbiased recommendation system that prevents the repeated assignment of the same employees. Building on this, Zhao et al. (2025) further argue that the single focus on utility goals proves to be inadequate in addressing real world concerns, and that the growing attention to fairness aware recommender systems is necessary to truly serve all the users of the system, which identifies the second need of the study, the need for a recommender system that considers both fairness and diversity in its suggestions so that no employee is consistently overlooked or overloaded. This is further supported by Rampisela et al. (2023), who points out that recommendation engines risk reinforcing existing biases by consistently overlooking qualified but underutilized

personnel, and with this the system will not recommend only the visible people, leaving others behind, which gives the third need of the study, the need for a provider fairness mechanism that ensures all qualified employees have an equal chance of being recommended regardless of how often they have been selected in the past. In the context of WEA, this is especially relevant because the company mentioned that the same people always end up being assigned to multiple projects at the same time because the leads do not have a system to track who is already loaded.

These systems are also essentials for business efficiency. A balanced system must help both the company and employee by distributing the workload more evenly, that are pointed out by Zhou et al. (2024), instead the manager are relying on the gut feeling or memory, a data-driven system approach that actual looks at availability. With this the fourth need of the study is identified, which is the need for a workload tracking system that distributes tasks evenly based on actual availability rather than personal preference or memory of the manager. Zhao et al. (2025) also identify that fairness and diversity in recommender systems are strongly connected, meaning that a system that is fair in its recommendations is also more likely to give diverse and balanced suggestions that will not keep repeating the same candidates, and this gives the fifth need of the study, the need to combine fairness, diversity and workload awareness into one recommendation logic instead of treating them as separate features. These findings gives a strong foundation in addressing the workload problem of this

study, showing the only way to truly identify the underutilized talent is through system that can analyzes employee's skills and current tasks without bias.

Synthesis 1

The literature consistently shows that the core problem of WEA is not just a matter of inefficiency but also of fairness. Jin et al. (2023) and Li et al. (2022) establish that a good recommender system must give equal chance to all employees, while Zhao et al. (2025) strengthen this by arguing that fairness and diversity in recommendations are strongly connected and must both be considered in the system design. Rampisela et al. (2023) further confirm that without a systematic and objective filter, recommendation engines will keep reinforcing the same biases that WEA is currently experiencing, and the workload problem identified through Zhou et al. (2024) shows that distributing tasks evenly based on actual availability is a necessary condition for a fair system. With all of these five needs justify that the proposed Automated Skill Extraction and Workload-Based Resource Recommendation System for WEA must be grounded in fairness, diversity and workload awareness to truly solve the problem of repeated and unbalanced employee assignment.

Lastly, switching to automated system can improves employee's morale and productivity. Bai et al. (2022) found that work is being assigned by a fair algorithm make the employees often feels more satisfied because the process is transparent and unbiased, and with this the sixth need of the study is identified, which is the need for an automated system that improves employee morale

through transparent and fair assignment process. Building on this, the seventh need is identified through Lima et al. (2024), who argue that automated and fair resource management is no longer optional but a strategic necessity for companies that want to retain skilled workers and maintain productivity, since the manual process of tracking who is actually free is making it hard, and this type of technology directly address that problem by removing the subjectivity from the assignment process. This is further supported by Banerjee et al. (2023), who point out that a fair recommender system must consider not just the company's needs but also the individual employee's stake in the assignment process, because a system that only optimizes for project completion without considering employee availability and satisfaction will eventually lead to burnout and disengagement, which gives the eighth need of the study, the need for a recommender system that balances company output with employee wellbeing.

Transitioning from this individual perspective to a broader organizational view, Zhou et al. (2024) support the ninth need of the study, which is the need for a sustainability aware recommendation approach that considers the long term impact of resource allocation decisions on both the company and its employees, instead the manager are relying on the gut feeling or memory, a data driven system approach that actual looks at availability and long term sustainability of the workforce. And this strategic shift are already in the local landscape, a study by Cacatian et al. (2025) AI integration in Metro Manila's HR recruitment shows that automated innovation is being a necessity to remain competitive, and with this the tenth need of the study is identified, which is the need for WEA to align

itself with the regional trend of adopting AI driven human resource tools to remain competitive in the Philippine industrial market. With these studies justify that adopting recommendations engine, WEA place itself with the regional trends, that moves toward to modernized and efficient resource allocation framework.

Synthesis 2

The remaining studies in this topic consistently shows, the literature consistently shows that the shift from manual to automated resource allocation is not just a technical upgrade but a strategic and cultural necessity. Bai et al. (2022) and Lima et al. (2024) establish that employee morale and organizational productivity are directly affected by the fairness and transparency of the assignment process, while Banerjee et al. (2023) and Zhou et al. (2024) further confirm that a truly effective recommender system must consider both the short term project needs and the long term sustainability of the workforce. And the local evidence from Cacatian et al. (2025) shows that this shift is already happening in the Philippine context, making it even more urgent for WEA to adopt this technology. With all of these five needs justify that the proposed Resource Recommendation System is not only technically necessary but also strategically and culturally aligned with the direction that modern organizations are taking in managing their human resources more fairly, transparently and efficiently.

Skill Matching and Employee-Project Alignment

Matching the right employee to the right project is one of the important part in making the resource management work properly. The process of identifying who is qualified for a specific task is not just about checking if the employee has the skill, but also making sure that the skill matches the exact requirement of the project. According to Mueller and Schaaff (2025), the AI based system for matching employees to project roles must not only consider the current skill level of the employee but also the interest of the employee because interest are an important motivator for workplace learning, and with this the first need of the study is identified, which is the need for an automated employee project matching system that goes beyond just checking skills and also considers the overall profile of the employee to produce better and more accurate assignments. Building on this, the second need is identified through Tang et al. (2025), who points out that person job recommendation systems are widely used in human resource management but many of them are considered as black box systems because their internal decision making processes remain impenetrable to the users, and this is very relevant to WEA because the team leads and the resource manager will only trust and use the system if they can clearly understand the basis of every recommendation, making transparency and explainability a necessary requirement for the proposed system.

The third need of the study is identified through the findings of Boukhari et al. (2024), who discuss that the skill matching at scale requires a system that can handle different types of candidate profiles and project descriptions in an efficient

way, and this is relevant to WEA because even if the company currently have a small team, the system must still be designed in a way that it can handle more employees and more projects as the company grows. Building on this, Kashif and Kumar (2024) found that using NLP and machine learning to extract resume content and analyze it can provide actionable feedback and improve the accuracy of matching candidates to the right roles, which support the fourth need of the study, the need for an NLP based resume extraction system that can automatically process employee documents and produce structured and searchable skill profiles for matching. Lastly, the fifth need is identified through Kishore and Sreerala (2025), who found that AI powered job matching systems that uses NLP and large language models can minimize the manual effort of reviewing resumes and improve the accuracy of matching candidates to the right positions, showing that technical skills, experience and job keywords can be extracted from resumes to allow more accurate candidate profiling that the system can use to compare against the project requirements.

Synthesis 1

What the literature reveals about skill matching and employee project alignment, the literature consistently shows that the core problem of WEA in matching employees to projects is not just a matter of inefficiency but also of accuracy and transparency. Mueller and Schaaff (2025) establish that a good matching system must go beyond just checking skills and must also consider the full profile of the employee, while Tang et al. (2025) strengthen this by arguing that transparency in the recommendation process is a critical requirement for gaining the trust of

the users. Boukhari et al. (2024) further confirm that the system must be scalable and capable of handling different types of documents and profiles, while Kashif and Kumar (2024) and Kishore and Sreerala (2025) both prove that NLP and machine learning based extraction is a reliable and accurate alternative to the manual review that WEA is currently doing. With all of these five needs justify that the proposed Automated Skill Extraction and Workload-Based Resource Recommendation System for WEA must be grounded in automated matching, transparent recommendations and automated skill extraction to truly solve the problem of slow and inaccurate employee project assignment.

The way skills are being represented and ranked in the system also affect how the final recommendation will turn out especially when multiple employees qualify for the same project. Naveed et al. (2024) found that using Natural Language Processing with similarity measures like Jaccard Similarity can extract the competencies of the employees from their resume and match them to predefined skill sets associated with the project requirements, and this is very relevant to WEA because the team leads mentioned that when two people can both do the task, they consider who has more relevant skills and more experience before making the final assignment, which gives the sixth need of the study, the need for a skill overlap and similarity measurement mechanism that can rank employees based on how closely their extracted skills match the specific requirements of the project. Building on this, the seventh need is identified through Decorte et al. (2024), who discuss that evaluating the relatedness between skills using self supervised learning can improve how the system understand the connection

between different technical skills even when they are expressed in different ways, which is a common situation in WEA where the employees submit their documents in different layouts and file types.

The eighth need of the study is identified through the findings of Clavié and Soulié (2023), who found that large language models can be used as zero shot skill matchers that can identify and match skills from job descriptions and candidate profiles even without requiring extensive training data, so the system can handle the technical and specialized skills that are common in the electrical distribution industry of WEA, like lighting calculations, junction box installations, and safety certifications, even when these skills are expressed in non standard terms or abbreviations. Transitioning from this technical skill matching perspective, the ninth need is identified through Magron et al. (2024), who discuss that generating synthetic and structured job postings and skill profiles can enhance the accuracy of skill matching, which is relevant to WEA because the company currently do not have a structured way of storing and comparing the skills of their employees. Lastly, the tenth need is identified through Gavrilescu et al. (2025), who found that using keyword extraction and transversal skill classification from job advertisements can help the system identify and organize the specific technical skills and competencies required for different types of projects, making it easier for the resource manager to see at a glance who is qualified and who is not without having to manually check every document.

Synthesis 2

Taking a closer look at the remaining studies in this topic, the literature consistently shows that the shift from manual to automated skill matching is not just a technical upgrade but a practical necessity for organizations like WEA that handles technical and safety critical projects. Naveed et al. (2024) and Decorte et al. (2024) establish that similarity measurement and skill relatedness evaluation are essential components of an accurate matching system, while Clavié and Soulié (2023) further confirm that even specialized and non standard technical skills can be matched accurately using modern language models without requiring large amounts of training data. And the findings of Magron et al. (2024) and Gavrilescu et al. (2025) shows that structuring and classifying skills in a consistent way is the key to making the matching process reliable and scalable as the company grows. With all of these five needs justify that the proposed Resource Recommendation System must be equipped with a robust and automated skill matching engine that can handle different document formats, recognize technical and specialized skills, and produce ranked and transparent recommendations that the resource manager can understand, verify and confidently act upon.

Optical Character Recognition (OCR) with NLP in Recommender Systems

Text recognition and extraction technology integration is one of the major step making recommendations in system are more efficient and automated. Combining Optical Character Recognition (OCR) with the personalized

algorithms allows system to process information and user to make more informed decisions, according to Zhang et al. (2023), which is important in organizations like WEA, where employee qualifications and certification must be verified quickly and accurately, and manual data entry is often time consuming and prone to error. With this the first need of the study is identified, which is the need for an automated document processing system that can convert different types of employee documents like scanned resumes and certificates into a readable and searchable text that the system can use for matching and recommendation. Building on this, Helli et al. (2023) found that combining OCR with NLP models to extract information from resumes after preprocessing can significantly improve the accuracy and efficiency of the information extraction process, showing that the automated document processing is reliable enough to replace the time consuming manual review of the employee certification and resumes that the team leads are currently doing in WEA.

The second need of the study is identified through the findings of Najam and Faizullah (2023), who shows that modern deep learning architectures can achieve the error rate as low as 2.03% in recognizing and extracting text from documents, which support the need to reduce delays in reviewing employee documents especially in WEA where the review of certifications and qualifications must be done before every project assignment. Transitioning from this accuracy perspective, Thangam et al. (2023) further support the third need of the study, which is the need for a post processing mechanism that can clean and improve the raw text output of the OCR system so that the NLP component can still

process it correctly even when the extracted text contains errors, noise or inconsistencies, which is a real situation in WEA because the employees submit their documents in different formats and quality levels. Malashin et al. (2024) report how the adaptive system can adjust themselves based on the document's layout to improve the extraction, giving the fourth need of the study, the need for an adaptive OCR system that can automatically adjust to different document layouts and formats. Lastly, Alsaif et al. (2022) found that advanced NLP techniques do not need perfect OCR text to successfully match jobs to candidates, and this is supported by Tarnowska and Ras (2021), whose NLP based system reached 88% accuracy in turning raw text into helpful business recommendations, which gives the fifth need of the study, the need for an NLP based text processing system that can convert the raw and unstructured text extracted by the OCR into meaningful and structured information.

Synthesis 1

The studies gathered on OCR and NLP document processing collectively shows, the literature consistently shows that the core problem of WEA in processing employee documents is not just a matter of speed but also of accuracy and reliability. Zhang et al. (2023) and Helli et al. (2023) establish that combining OCR with NLP is a reliable and efficient approach for automatically converting employee documents into structured and searchable information, while Najam and Faizullah (2023) strengthen this by proving that modern OCR systems can achieve very low error rates that are more than acceptable for a professional environment like WEA. Thangam et al. (2023) and Malashin et al. (2024) further

confirm that even when the document quality is inconsistent, the system can still produce accurate results through post processing and adaptive extraction techniques, and Alsaif et al. (2022) and Tarnowska and Ras (2021) both prove that the NLP component can still match qualifications accurately even when the extracted text is not perfectly clean. However, a difference can be noted between Malashin et al. (2024) who focus on layout adaptation and Thangam et al. (2023) who focus on post processing correction, showing that the two approaches address the same problem from different angles and both must be considered in building a robust document processing pipeline for WEA. With all of these five needs justify that the proposed Automated Skill Extraction and Workload-Based Resource Recommendation System for WEA must be grounded in a robust and automated OCR and NLP pipeline that can handle different document types and qualities to truly solve the problem of slow and inconsistent manual document review.

Lastly, WEA currently has a small team and the leads already know the skills of each person informally, the OCR and NLP features become more important as the company grows or when new employees are onboarded. Janani et al. (2025) found combining OCR with machine learning algorithms like K-Nearest Neighbors can produce more accurate profiles, which gives the sixth need of the study, the need for a scalable document processing system that combines OCR with machine learning algorithms so the system can produce more accurate and complete employee profiles as the number of employees and documents increases over time. Building on this, Gaur et al. (2021) used the Named Entity

Recognition or NER model to parse unstructured data into searchable profiles, identifying the technical skills and certifications from the resumes even when the documents are not in a standard format, which gives the seventh need of the study, the need for a Named Entity Recognition component that can specifically identify and extract the technical skills, certifications and experience levels from the unstructured text. Sharma et al. (2023) found that hybrid methods combining TF-IDF and Word2Vec can achieve up to 92.7% accuracy in matching skills from resumes to job requirements, which is more than enough to replace the manual review process that the team leads are currently doing, giving the eighth need of the study.

Transitioning from this technical accuracy perspective, the ninth need of the study is related to the overall impact of combining all of these technologies together in one system. Zhang et al. (2023) discuss that the integration of OCR with personalized recommendation algorithms allows the system not only to extract information but also to use that information to make more informed and accurate decisions, giving the ninth need, the need for an integrated OCR and recommendation system that can extract, process and directly use the information in the recommendation engine. Lastly, Helli et al. (2023) also found that classifying text groups from OCR processed resumes using NLP models can produce structured and organized employee profiles that the system can store and use for future matching, giving the tenth need of the study, the need for a structured employee profile storage system so the team leads does not need to

re upload and re process the documents every time a new project assignment is needed.

Synthesis 2

The literature consistently shows that the shift from manual to automated document processing is not just a technical upgrade but a strategic investment in the long term operational capacity of WEA. Janani et al. (2025) and Gaur et al. (2021) establish that combining OCR with machine learning and Named Entity Recognition is essential for producing accurate and complete employee profiles that the system can use for matching and recommendation, while Sharma et al. (2023) further confirm that hybrid NLP methods can achieve accuracy rates that are more than sufficient to replace the manual review process that the team leads are currently doing. But a contradiction can be noted between the findings of Janani et al. (2025) who focus on K-Nearest Neighbors for profile building and Gaur et al. (2021) who focus on Named Entity Recognition for skill extraction, which shows that there is no single best approach and the proposed system must combine both methods to achieve the most accurate and complete employee profiles. Zhang et al. (2023) and Helli et al. (2023) both prove that integrating OCR with recommendation algorithms and structured profile storage can transform the document processing pipeline into a fully automated and scalable system that WEA can rely on even as the company grows and onboards new employees in the future. With all of these five needs justify that the proposed Resource Recommendation System must be equipped with a complete and automated OCR and NLP pipeline that can extract, process, classify and store

the information from different types of employee documents, so that the system can produce faster, more accurate and more reliable employee assignment recommendations that the team leads can trust and confidently act upon.

Rapid Application Development (RAD) for Efficient System Deployment

Rapid Application Development or RAD is now well known and recognized as an effective software development methodology because it gives fast prototyping making the work more easily and efficient to the users. According to Novita et al. (2023) and Meyliana et al. (2024), it shows that Rapid Application Development helps produce successful systems while reducing the development risks, particularly when users are involved early in the design and testing stages, and with this the first need of the study is identified, which is the need for a development methodology that can involve the users of WEA early in the design and testing stages so that the system can be adjusted based on the actual needs of the team leads before the final version is completed. Building on this, Parlika et al. (2023) further support by showing that RAD is consistently identified as the most efficient model in software development because it has the shortest processing time compared to other methodologies, giving the second need of the study, the need for a development methodology that can produce a working system in the shortest possible time so that the team leads of WEA can start using the workload tracking and recommendation features as soon as possible. The third need of the study is identified through the findings of Brill (2025), who explains that RAD is an iterative approach that focuses on quick functional prototypes, regular user feedback and flexibility in changing requirements,

making sure that the final version of the system truly reflects the actual workflow and assignment process that WEA is currently using.

Building on this, Riadi et al. (2024) found that the RAD method has a significant positive impact on both the development cycle and user satisfaction, particularly in web based systems where the users need to interact with the system regularly, giving the fourth need of the study, the need for a development methodology that can ensure a high level of user satisfaction so that the team leads of WEA will actually use the system consistently after it is deployed instead of going back to the manual process. Lastly, the fifth need is identified through Protasiewicz et al. (2021), who shows that the adaptability of RAD allows it to be applied not only in information systems but also across various domains such as financial applications and inventory management too because of the flexibility of Rapid Application Development, supporting the development of the Automated Skill Extraction and Workload-Based Resource Recommendation System for WEA, as the project requires rapid development and the ability to adapt features based on both company requirements and user needs.

Synthesis 1

Building on the earlier findings in this topic, the literature consistently shows that Rapid Application Development is the most appropriate methodology for building the proposed Resource Recommendation System. Novita et al. (2023) and Meyliana et al. (2024) establish that RAD reduces development risks and produces successful systems by involving the users early, while Parlika et al.

(2023) strengthen this by proving that RAD is the most efficient model in terms of development time. Brill (2025) and Riadi et al. (2024) further confirm that the iterative and prototype driven nature of RAD leads to higher user satisfaction and better alignment between the system and the actual needs of the users. But a difference can be noted between Riadi et al. (2024) who focus on user satisfaction and Parlika et al. (2023) who focus on development efficiency, which shows that RAD is not just about speed but also about making sure the users are happy with the system that is being built. With all of these five needs justify that adopting RAD as the development methodology of the proposed system is not just a practical choice but a strategically sound decision that directly addresses the development constraints and the business requirements of WEA.

But some studies argue that Rapid Application Development focuses too much on speed, which can sometimes affect the scalability and maintainability of the system in the long run (Carnovalini, 2025), and without proper planning this can cause a problem in the future even if RAD is useful for developing systems. With this the sixth need of the study is identified, which is the need for a careful and well documented development plan that complements the rapid pace of RAD to ensure that the proposed system remains reliable and scalable even after deployment. Building on this, Wang et al. (2025) discuss that functional prototypes produced through RAD can help developers detect and fix problems early, but only if the testing process is thorough and well planned, giving the seventh need of the study, the need for the system to be tested carefully at every iteration to ensure that the OCR, NLP and recommendation components are all

working correctly before they are integrated into the final system, which is very relevant to WEA because any error in the recommendation output could directly affect the operations of the company.

Lastly, the eighth need is identified through Imran and Bhutto (2023), who found that RAD based web application development is particularly effective for small and medium organizations because the methodology is designed to work well with small development teams and limited resources, meaning the system must be developed in a way that is realistic and manageable for a small research team building a system for a small company like WEA. The ninth need is identified through Stankovic (2024), who discuss that RAD in modern software engineering is still relevant and effective especially for projects that requires fast delivery and continuous user involvement. And the tenth need is identified through Carnovalini (2025), who warns that the risk of prioritizing speed over quality in RAD must be managed through careful planning and documentation.

Synthesis 2

Together, the literature consistently shows that the success of RAD as a development methodology is depends not just on the speed of development but also on the quality of planning, testing and user involvement throughout the entire development process. Carnovalini (2025) warns that without proper planning the speed of RAD can become a liability rather than an advantage, while Wang et al. (2025) and Imran and Bhutto (2023) both confirm that thorough prototype testing and a development approach that matches the size and

capacity of the organization are the key factors that will determine whether a RAD based system will perform reliably in a real world environment like WEA. But a difference can also be noted between Stankovic (2024) who focus on the relevance of RAD in modern software engineering and Carnovalini (2025) who focus on the risks of prioritizing speed over quality, which shows that RAD must be applied carefully to avoid long term problems especially for a system that will be used for critical project assignment decisions in WEA. With all of these five needs justify that the proposed Resource Recommendation System must be build using a well planned and carefully documented RAD process that balances speed with quality and rapid prototyping with thorough testing to ensure that the final system is not only built quickly but also built correctly and reliably for the long term use of WEA.

Evaluating System Effectiveness

To see if the transition from manual to automated resource management is successful, the system must be measure against the international standards. The ISO/IEC 25010 standard will serve as the standard framework in evaluating that will allows researchers to analyze both the technical functionality and user satisfaction (Nugraha and Rachman, 2024), and by applying this standard, the study can move beyond subjective opinions and provide the assessment on how the system performs in real word corporate environments. With this the first need of the study is identified, which is the need for a standard evaluation framework that can measure how well the proposed system works compared to the manual process that WEA is currently doing. Building on this, Rojas et al. (2025) shows

that ISO/IEC 25010 has been consistently used across different domains and industries which proves that the standard are flexible and reliable enough to be apply in different organizational context including the resource management context of WEA, giving the second need of the study, the need for an evaluation framework that are not just technically rigorous but also widely accepted across different fields so that the findings of the evaluation can be considered credible.

There's literature that demonstrates high performance in the quality areas is achievable for institutional systems. Kasri et al. (2024) found that functional suitability and reliability score as high as 98% and specialized information system as 100%, giving the third need of the study, the need for the proposed system to achieve a high level of functional suitability and reliability that are comparable to the benchmark scores in similar institutional systems. Similar to Nuzula and Rochimah (2023) that found human resource management tools can achieve 90% in appropriate and completeness in performance, which suggest different quality characteristics are interconnected, giving the fourth need of the study, the need for the proposed system to perform consistently well across all the quality characteristics because a weakness in one area can affect the overall performance of the system. Lastly, Barletta et al. (2023) highlight that evaluating factors such as safety, response time and context help demonstrate the quality from the end user's point of view, giving the fifth need of the study, the need for the proposed system to be evaluate not just from a technical perspective but also from the perspective of the actual users of WEA who will be using it on a daily basis.

Synthesis 1

The further studies in this topic strengthens, the literature consistently shows that the ISO/IEC 25010 standard is the most appropriate framework for evaluating the quality of the proposed Automated Skill Extraction and Workload-Based Resource Recommendation System for WEA. Nugraha and Rachman (2024) establish that ISO/IEC 25010 provides a comprehensive framework for measuring both technical functionality and user satisfaction, while Rojas et al. (2025) strengthen this by showing that the standard are widely used and validated across different domains. Kasri et al. (2024) and Nuzula and Rochimah (2023) further confirm that high performance scores are achievable for institutional and human resource management systems that provides a meaningful benchmark for the proposed system to aim for. But a difference can be noted between Kasri et al. (2024) who focus on the functional suitability and reliability and Barletta et al. (2023) who focus on the quality from the end user's point of view, which shows that evaluating the system from both the technical and the user perspective is necessary to get the complete picture of the system quality. With all of this studies justify that the evaluation of the proposed Resource Recommendation System must be grounded in ISO/IEC 25010 and must cover both the technical performance and the user experience to see if the system is really a better alternative than the manual process that WEA is currently struggling with.

The technical reliability are only one part of the equation, what truly defines the system's impact is the quality of use from the human perspective. Gotz (2023)

discuss that system with smaller datasets, a simpler recommendation can be highly effective without requiring decades of data history, giving the sixth need of the study, the need for the proposed system to still produce reliable recommendations even if WEA do not have a long record of past project assignments since the company only relies on Excel files for tracking. Building on this, De Campos et al. (2024) and Alberto et al. (2025) both found that a system that gives short, clear and explainable suggestions is the key for gaining the confidence of the resource manager, giving the seventh need, the need for the proposed system to give clear explanation for every recommendation it makes so the team leads of WEA can understand why a certain employee is being suggested and can act on it without hesitation.

Lastly, the eighth and ninth needs are identified through the findings of Ilyas and Ahmad (2023) and Susanto and Andriana (2023), who both show that usability and functional suitability are the most critical quality characteristics especially for systems that are being used by non technical users, and that measuring the quality using ISO/IEC 25010 can help identify specific areas where the system still needs improvement before it is officially deployed. This is very relevant to WEA because the team leads who will be using the system are not IT professionals, and the system must be easy to navigate and understand for them to actually use it in their daily assignment process without needing technical support. And the tenth need is that the system must follow the ISO/IEC/IEEE 26514:2022 standard that ensure the user manuals and troubleshooting guides are clear and user friendly.

Synthesis 2

The studies reviewed on system evaluation consistently shows, the literature consistently shows that the quality of use from the human perspective is just as important as the technical performance of the system. Gotz (2023) and De Campos et al. (2024) both shows that even with small and limited data, the system can still produce useful and trustworthy recommendations if the suggestions are clear and directly connected to what the user actually needs, while Alberto et al. (2025) further support this by showing that the clarity and relevance of the recommendations are what makes the users actually accept and use the system in their daily work. But a difference can be noted between Ilyas and Ahmad (2023) who focus on usability and functional suitability as the most important quality characteristics and Susanto and Andriana (2023) who focus more on using the evaluation results to find and fix the weak areas of the system before deployment, which shows that the ISO/IEC 25010 evaluation is not just a one time judgment but also a tool for making the system better. With all of this studies justify that the proposed Automated Skill Extraction and Workload-Based Resource Recommendation System for WEA must be evaluate in a way that covers both the technical quality and the practical usability of the system so that the team leads of WEA can truly rely on it every day to make faster, fairer and more accurate project assignment decisions than the manual process they are currently doing.

Significance of the Study

The results of this study will benefit WEA by providing automated approach to managing workforce allocation in the industrial distribution sector. The implementation of the Resource Management Recommender System, the company will able to transition from slow manual review process to a data driven method in assigning the right personnel to the right tasks.

For the organization (WEA), the system will help manage the technical workforce more effectively by reducing the time spent on manual checking of internal resume and certification. This improvement minimizes the human error, enhance the safety compliance and support the company's goal of maintaining the high quality of project delivery.

For the resource and project manager, the system made daily operation more manageable. They will be able to quickly identify the qualified employees who met the strict technical requirements of each project, that prevent scheduling conflicts and operational delays. This allow managers to make faster, more accurate decisions and especially when the organization grew and handle multiple number of complex projects.

For employees, the system will provide transparency and fair work environment. Their skills and qualifications will recognize accurately through the system ability to analyze different documents formats that ensure task assignments is based on the actual availability. The focus on verifying

certification ensure that employees are only assign to roles they are qualified for, increasing their professional confidence and safety in the job.

For the researchers, the study will serve as opportunity to enhance their technical and competencies. Developing a system that integrates complex AI tools like OCR and NLP, the researchers will able to apply theoretical programming knowledge to solve the real world industrial problems. With this project, it allows them to gain deeper understanding of the human resource management and safety standards of oil and gas industry that will prepare them for professional careers in software development and project management.

Lastly, the future researcher, this study will serve as useful reference for someone who are interested in exploring AI driven human resource management. The integration of the OCR and NLP for specialized industrial certification will provide foundation for developing similar intelligent system in other fields. Future developments can build by remodel the system to include features like automated attendance, payroll integration or real time tracking of license expirations.

Statement of the Problem

The success of WEA depends on the Resource Manager's ability to assign the employees to the specific requirements in a project and ensure that the workforce resources are being used effectively. The company's Resource Manager are currently manually checking the resumes and certifications that

make it slow and can lead to human mistakes and it hard to see if the employee is actually available. In this approach it consumes time, is prone to human error and makes it more difficult to determine if the employee possesses the required competencies or is already assigned to multiple projects. As the result, some of the employees may become overloaded with the project assignments while other qualified personnel remain underutilized. To solve this challenges, the proposed An Automated Skill-Extraction and Workload-Based Resource Recommendation System it aims to automate skills and certifications extraction, monitor employee workload and availability and provide recommendations for the project assignments based on the employee qualifications, experience and and current resource utilization.

Specifically, the study seeks to answer the following research questions:

1. What functionalities can the proposed An Skill-Extraction and Workload-Based Resource Recommendation System provide to automate the processing of employee internal resume and certifications?
2. What functionalities can the proposed An Skill-Extraction and Workload-Based Resource Recommendation System provide to assist Resource Managers in identifying employees whose skills, experience and certifications meet project requirements?
3. What recommendation capabilities can the An Skill-Extraction and Workload-Based Resource Recommendation System provide to assist Resource

Managers in identifying the most suitable and available employees for project assignments?

4. What is the level of acceptability of the proposed An Skill-Extraction and Workload-Based Resource Recommendation System in term of:

- a. Functional Suitability
- b. Reliability
- c. Usability

Scope and Limitation of the Study

The focus of this study is the development and evaluation of the Resource Management Recommender System designed for WEA, a small electrical distribution company with fourteen to fifteen employees that operates without a local Human Resources department in the Philippines. The company is being managed remotely by their main office in Singapore, and the assignment of employees to projects is being handled by the team leads. The system core functions is center on two primary goals. First is the workload tracking feature that will allow the team leads and the Singapore manager to see the current availability and workload status of each employee in real time. Second is the automated skill extraction using Optical Character Recognition or OCR, a technology used to convert different types of documents like scanned paper or PDFs into editable and searchable text, and Natural Language Processing or NLP, a field of artificial intelligence that enables computer to understand, interpret and process human language. With combining these two technologies, the

system will automate the extraction and interpretation of skills from the internal resume and technical certificates of the employees.

The scope of the study is to design the system using the Rapid Application Development or RAD model to allow iterative design, prototyping and user feedback. The system are used to process documents in PDF and common image formats such as JPG and PNG that are the commonly used formats for engineering credentials and internal resumes. The study is also limited to the context of WEA only, which means the system is design specifically to match the workflow and the assignment process that WEA is currently using. To ensure the privacy and confidentiality of the personnel records. The system use synthetic datasets that are simulated records that mimic the characteristics of the real world resume and certification without exposing sensitive personal information of the actual WEA employees. The system performance is also evaluated using the ISO/IEC 25010 software quality standard, that focus on the functional suitability, usability and reliability.

This study has several limitations. The accuracy of the OCR component is highly dependent on the document quality, because of this handwritten notes or low resolution scans are outside the system's processing capabilities. The NLP component is enhanced for English language documents and may struggle with highly specific regional jargon and non standard abbreviations that are commonly used in the electrical distribution industry. Due to the compliance with the Data Privacy Act of 2012, the actual company personnel records from WEA are not utilized in the testing phase to protect the confidentiality of the employees.

Professional licenses are also not a primary matching criterion for most of the WEA projects, with the exception of safety certifications that are only required for specific on-site work such as the UPS maintenance in Petron Refinery. The system also does not cover the management of the Singapore office directly, but it will provide a dashboard that the Singapore manager can access remotely to see the workload and availability of the Philippines team. Lastly, the system is designed as a web-based platform that requires an active internet connection for data processing and does not cover broader business functions like payroll processing, daily attendance tracking or financial accounting.

The system also relies on the historical project records of the employees to determine the experience weight or E value that is used in the recommendation ranking formula. Since WEA does not currently have a structured digital record of past project assignments, the initial experience data of the employees will be encoded manually by the team leads during the system setup phase. This means that the accuracy of the E value is dependent on the completeness and honesty of the information that is provided during the initial data entry. For the new employees who do not have any recorded past project experience in the system, the E value will default to 1 to ensure that they are still included in the recommendation results and will not be completely ignored by the ranking algorithm. The task weight values that are used in the workload score formula are also predefined by the researchers in coordination with the team leads of WEA to ensure that the weight assigned to each task type reflects the actual complexity and demand of the work in the electrical distribution industry.

CHAPTER 2

METHODOLOGY

Research Design

The study use descriptive-developmental research design because it focuses on both building technological solution and business problem. According to Meyliana et al. (2024), a developmental research design is a systematic process that focuses on designing, developing, and evaluating products to ensure they are effective and consistent. In the other hand, the descriptive research part of this design allows researchers to study and explain the current manual inefficiencies at the company, and the developmental part focus in the actual creation and testing of the Resource Management Recommender System. The system is built using Rapid Application Development methodology, Meyliana et al. (2024) explained it is an iterative process that focuses on fast prototyping and user involvement to speed up development.

This design is most suitable choice because it provide clear path to identify the company's specific needs first and then develop the OCR and NLP tools that required to solve the real world delays and errors identified in the study.

Architectural Framework

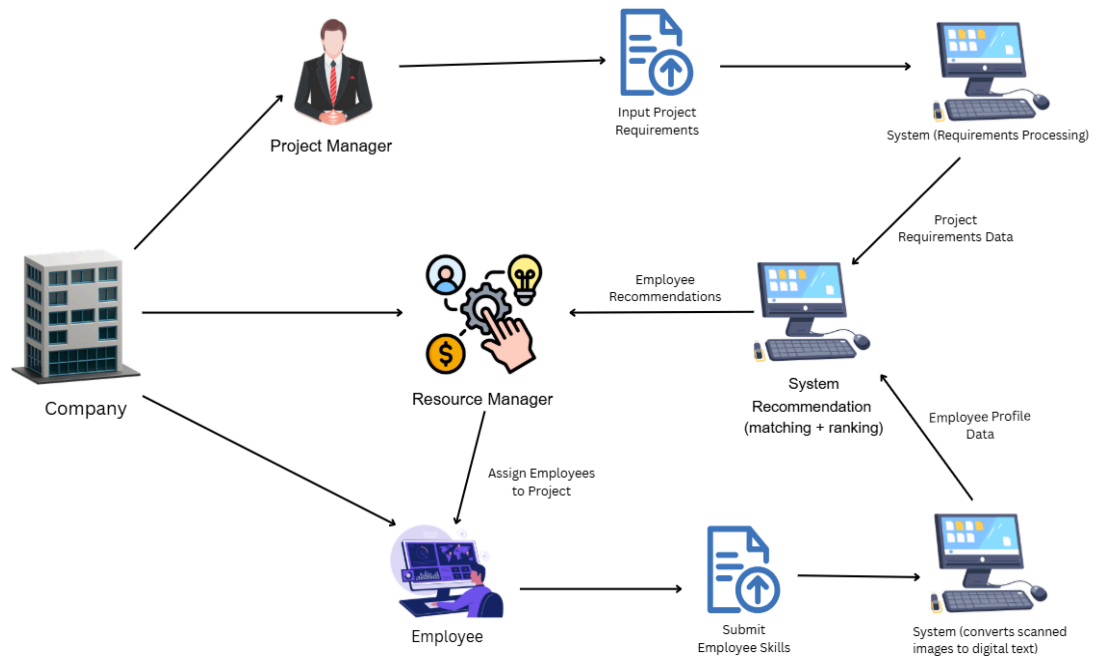


Figure 1. Architectural Framework

Figure 1 shows the architectural framework of the Resource Management Recommender System. It explains the flow from the Company's Project by Manager inputting project requirements and Employees submitting their skill documents, both data flow being processed by the System through requirements processing and OCR conversion respectively, the System Recommendation engine performing matching algorithms using the processed data, and finally the Resource Manager receiving employee recommendations to assign the most suitable employees to the projects.

Level 0 DFD/Schematic Diagram

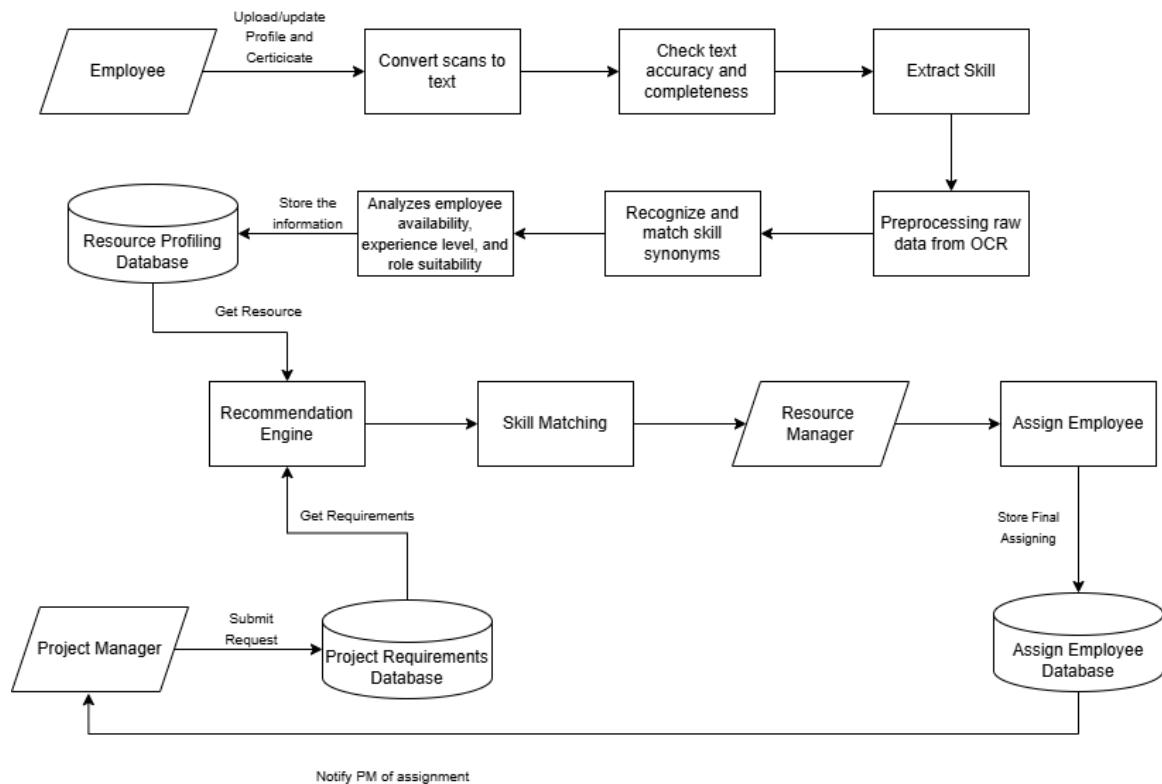


Figure 2. Data Flow Diagram (DFD)

Figure 2 shows the Data Flow Diagram (DFD) for the Resource Management Recommender System. The diagram explains how data moves in the system, it start with from employees submitting their resumes and certificates, to evaluating and assigning them to projects. The system check the documents are correct, extracts and verifies employee information and skills, and matches the skills to project requirements

Algorithm Discussion

Skill Extraction Representation

The extracted skills from the documents of the employee are represented as a set where S_e is the set of employee skills and S_p means set of required project skills. For example a project for WEA might require a set = {Electrical Distribution, Safety Certification, Transformer Installation}. The Optical Character Recognition or OCR converts the documents from employees into a readable machine text, while Natural Language Processing or NLP processes the text to extract the meaningful skills. These skills are going to be normalized through synonym mapping and token analysis to ensure the consistency of the different formats of employee data.

Skill Matching Function

The system will compute the number of matched skills using set intersection, represented as $M = |S_e \cap S_p|$. The overlapping skills between the project requirements and employees will be identify using this formula. The value of M would be 2 if an employee internal resume contains Electrical Distribution and Safety Certification out of the 3 required project skills. Using the Natural Language Processing or NLP, it enhance the processing by recognizing the related terms and jargons that will allow matching if the wordings are different.

Skill Matching Score

The matching score will be computed by dividing the M by the total number of needed required project skills |Sp|. Using the example in the previous step, the score is $2 \div 3$, resulting in 0.67. This will produce a normalized value between 0 and 1 representing how the employee meets the project requirements.

Experience Weighting

To prioritize the employees that have more relevant experience, the system applies a weighting factor represented as E. Unlike a fixed scale that is based on years of service, E is based on the number of relevant past projects that the employee has completed that are of the same type or category as the current project. This means that an employee who has completed more similar projects in the past will receive a higher E value and will be ranked higher when skills matches are equal. For example if an employee has completed 3 relevant past projects, E is 3. This approach is more accurate for WEA because the company evaluates the employees based on their actual hands-on experience in the specific project types rather than just the number of years they have been working.

Workload Score

To ensure that the system will not repeatedly assign the same overloaded employees to the new projects, the current workload of each employee is being measured using the formula $W = \sum (\text{Task Weight} \times \text{Task Count})$. The Task Weight

represents how demanding a task is, where a more complex task is given a higher weight, and the Task Count represents how many tasks of that type the employee currently has. The results are being added together to produce the total workload score W . The higher the value of W , the more loaded the employee is. This directly addresses the problem at WEA where the team leads tend to assign the same familiar employees repeatedly without knowing who is actually available.

Recommendation Ranking

Employees will be ranked based on their final score Re , which is computed using the formula $Re = \text{Matching Score} \times E \times 1/(1+W)$. This formula combines three factors that are the skill fit of the employee through the Matching Score, the relevant past project experience through E , and the current availability through $1/(1+W)$. The term $1/(1+W)$ ensures that the heavily loaded employees are automatically ranked lower even if they have strong skills and experience. The $+1$ in the denominator is necessary to prevent the division by zero in cases where the employee has no current workload, meaning $W = 0$. In that case, $1/(1+0) = 1$, so a fully available employee receives no penalty to their score. The system sorts all candidates in descending order of Re and presents the ranked list to the Resource Manager so that the most suitable and available employee is always recommended first.

Supplemental Figures

Mathematical Formulations for WEA Resource Management

Skill Extraction and Matching

$$M = |S_e \cap S_p|$$
$$\text{Matching Score} = \frac{|S_e \cap S_p|}{|S_p|}$$
$$\text{Workload Score} = \sum (\text{Task Weight} \times \text{Task Count})$$

Ranking and Allocation

$$R_e = \text{Matching Score} \times E \times \frac{1}{1 + W}$$

Variable and Parameter Key

Symbol	Description
S_e	Set of employee skills extracted using OCR/NLP
S_p	Set of required project skills
M	Total number of matched skills (Set Intersection count)
W	Workload score; higher W reduces recommendation priority
E	Experience weight based on number of relevant past projects completed
R_e	Final recommendation score (Ranking)

Figure 3. Mathematical Formulation and Variable Definitions

Sample Data

Category	Sample Extracted Data (Raw)	Normalized Token (System Output)
Job Role	Electrical Maintenance Technician	Technician
Technical Skill	Distribution Transformer Installation	Electrical Distribution
Safety Cert	OSH 40-Hour Training	Safety Certification
Past Projects	3 completed electrical distribution projects	3 Relevant Past Projects (E = 3)

Table 1. Sample Data Extraction Table

Algorithm Walkthrough

Step	Variable	Sample Calculation	Result
1. Skill Matching	M	Employee has 2 out of 3 required skills	2
2. Match Score	Score_match	$2 \div 3$	0.67
3. Experience Weight	E	Employee completed 3 relevant past projects	3
4. Workload Score	W	$(2 \text{ tasks} \times \text{weight } 3) + (1 \text{ task} \times \text{weight } 1)$	7
5. Final Ranking	Re	$0.67 \times 3 \times 1/(1+7) = 0.67 \times 3 \times 0.125$	0.25

Table 2. Step-by-Step Algorithm Walkthrough

SE Paradigm

Rapid Application Development (RAD) Methodology

This study will use the Rapid Application Development (RAD) methodology to create OCR and NLP based resource management system. RAD is iterative approach to focus on quick functional prototypes, regular user feedback, and flexibility in changing requirements. This methodology is good for this project because the OCR and NLP components need repeated testing and improvement using document and user input (Brill, 2025). This study follow phases:

Phase 1: Requirements Planning

In requirements planning phase, will identify the types of documents that system will be process using OCR and NLP. Also determining the information that will be extract from the uploaded documents. The development team should understand the system's expected functionality and objectives.

Phase 2: User Design

In user design phase, functional prototypes are develop for OCR text extraction, resource recommendation dashboards and document upload interface. The functional prototype will evaluate by respondents and review by the project requirements.

Phase 3: Rapid Construction

Iteration 1: OCR and Document Processing

The first iteration focus in developing the OCR component that extract texts from documents. The system will be testing using sample documents to ensure accurate and reliable text extraction. Every errors may found during the testing will be fixed before moving to the next iteration

Iteration 2: NLP and Information Extraction

In the second iteration, NLP will apply to extracted information to find and extract relevant information for the resource management. With repeated testing sample document, will ensure the system correct interpret and organize information.

Iteration 3: Resource Matching and Recommendation Algorithms

In last iteration involves in developing algorithms to match extracted resources with project needs and generate recommendations. The algorithms are will be test using sample data to verify their accuracy, relevance, and efficiency to see if it is providing meaningful suggestions.

Phase 4: Cutover

After all the iterations, the system parts are put together and tested to see if they work well. The system will be check using test cases and sample data to make sure it works correctly and performs okay before it is used.

Benefits of RAD

Using RAD helps the system faster because working functional prototypes can be developed quickly. Feedback from test cases and internal evaluations helps improve the system. This allows the NLP and recommendation algorithms to be change based on the results from early tests with this the system more accurate and reliable.

Population and Sampling

The target population of this study have key personnel that are in workforce allocation and technical operation within WEA. It includes the Resource Managers that are responsible for assigning employees to projects, Project Managers who decide the specific manpower and safety requirements for each hazardous industrial sites and the Skilled Employees whose qualifications and certifications serves as the primary data inputs of the system. Focusing in this roles, the study make sure that the proposed Skill-Extraction and Workload-Based Resource Recommendation System is reflects the actual operation processes and challenges that encountered in managing construction personnel.

The study use purposive sampling, a technique that used to select participants based on their specific relevance and expertise. This is appropriate because the evaluation of the proposed system is requires respondents with specialized knowledge in workforce allocation, construction project management and software development. According to Andrade (2021), purposive sampling is

effective when researchers intentionally choose participants who possess the qualifications and characteristics that are necessary to address the study's objectives. Total of fourteen (14) are purposively selected as the respondents who will evaluate the system based on the ISO/IEC 25010 software quality standard. This group is consist one (1) domain expert with experience in workforce documentation and validation to verify the business logic, three (3) technical experts that will assess the system's architecture and the integration of the OCR and NLP components and lastly the ten (10) end users who will evaluate the system usability and functionality within the WEA context. This sample size is justified because it provides balanced perspective from the both technical developer and actual business people who will utilize the system.

Data Collection Methods

The primary instrument used in this study is a structured Likert-scale questionnaire that is based on the ISO/IEC 25010 international software quality standard. The questionnaire was used to gather the evaluations and assessments of the fourteen (14) purposively selected respondents, which consists of ten (10) end users, three (3) technical experts, and one (1) domain expert. It is designed to assess the quality of the An Skill-Extraction and Workload-Based Resource Recommendation System in terms of functional suitability, usability, and reliability of the system. Each item or question in the questionnaire are corresponds to a specific quality characteristic and is rated

using a Likert scale so that the respondents are able to express the degree of their agreement on how the system performs.

The scale ranges from 1 to 5, where 1 means Strongly Disagree and 5 means Strongly Agree this is the way of allowing the evaluators to clearly rate the system quality characteristic based on their actual experience in using the system. The questionnaire is divided into sections that cover the three main quality areas of the study. The functional suitability section checks whether the system is able to correctly and function properly the extract skills from the documents, matching employees to projects, and produce accurate recommendations of employee to their specific project. It also includes indicators that are directly aligned with the research questions of the study. The first research question focuses on how OCR-based automation can reduce the delay in processing documents, so additional items were included to evaluate if the system can accurately categorize specific employee information such as certifications and expiry dates from different document formats. The third research question addresses the effectiveness of the recommendation engine in identifying underutilized employees, so an indicator measuring the system's ability to show the availability of employees not currently assigned to any project was also added. Lastly, to address the fourth research question regarding the impact of the system on the turnaround time for assigning employees, an item was included to assess whether the automated recommendation process significantly reduces the time the resource manager spends compared to manual checking.

The usability section evaluates if the interface is easy to navigate and understand for each type of user, which are the Resource Manager, Project Manager, and Employee. And the reliability section assesses if the system performs consistently and without errors during the normal use. The second section that was originally labeled as usability is now properly labeled as reliability to correctly categorize the indicators that measure the system's stability and error-handling behavior, which are consistent with the scope and limitations of the study. By using this instrument, the researchers can gather a structured and measurable data that will be used to determine how well the An Skill-Extraction and Workload-Based Resource Recommendation System meets the quality standards that is set by ISO/IEC 25010.

Pamantasan ng Lungsod ng Pasig – College of Computer Studies
Resource Management Recommender System
 ISO/IEC 25010 Software Quality Evaluation Questionnaire

Please rate each item based on your experience using the system. Use the scale below:

1 — Strongly Disagree 2 — Disagree 3 — Neutral 4 — Agree 5 — Strongly Agree

A. Functional Suitability

Indicator	1	2	3	4	5
1. The system accurately converts uploaded employee CVs and certificates into structured, searchable profiles.	☐	☐	☐	☐	☐
2. The system reduces the time needed to process and review employee documents compared to manual checking.	☐	☐	☐	☐	☐
3. The system correctly matches employee competencies and certifications to the specific requirements of a project.	☐	☐	☐	☐	☐
4. The system effectively identifies qualified employees from the WEA resource pool who meet the technical and safety requirements of the project.	☐	☐	☐	☐	☐
5. The recommendation engine helps the resource manager identify skilled employees who are currently available but underutilized.	☐	☐	☐	☐	☐
6. The ranked employee suggestions produced by the system are relevant and accurate based on the project's requirements.	☐	☐	☐	☐	☐
7. The system accurately categorizes information (e.g., name, specific certifications, expiry dates) from various document formats (PDF/Images).	☐	☐	☐	☐	☐
8. The system provides visibility into the availability of employees who are not currently assigned to any active projects.	☐	☐	☐	☐	☐
9. The automated recommendation process significantly reduces the time spent searching for qualified personnel compared to manual resume review.	☐	☐	☐	☐	☐

B. Usability

Indicator	1	2	3	4	5
10. The system interface is easy to navigate and understand for my role.	☐	☐	☐	☐	☐
11. The features available for each user role (Resource Manager, Project Manager, Employee) are clearly accessible.	☐	☐	☐	☐	☐
12. The document upload process is simple and easy to use.	☐	☐	☐	☐	☐
13. The recommendation results are presented in a clear and understandable way that supports faster assignment.	☐	☐	☐	☐	☐

C. Reliability

Indicator	1	2	3	4	5
14. The system performs consistently without errors during normal use.	☐	☐	☐	☐	☐
15. The system does not crash or produce unexpected errors when processing employee documents.	☐	☐	☐	☐	☐
16. The system responds within an acceptable time when assigning employees to a project, compared to the manual process.	☐	☐	☐	☐	☐

Comments / Suggestions (optional):

Figure 4. ISO/IEC 25010 Evaluation Questionnaire

Figure 4 provides the design of the questionnaire used in this study. It is designed specifically to gather the structured assessment of the evaluators on the different quality characteristics of the system based on the ISO/IEC 25010

standard. This same questionnaire is used as the instrument to collect all the data needed for evaluating the overall quality of the Resource Management Recommender System.

The data collection started with preparing the ISO/IEC 25010-based Likert-scale questionnaire that will be used by the evaluators in assessing the system. After completing the system functional prototype, the researchers let the purposively selected respondents use and interact with the features of the Resource Management Recommender System. The evaluators were given time to explore the system by performing tasks that are relevant to their role, such as uploading employee CVs and certificates through the Employee Portal, viewing the ranked employee recommendations on the Edit Project and Assign Team screen, and checking how the system match the employee skills to the project requirements. After using the system, the evaluators answered the ISO/IEC 25010-based Likert-scale questionnaire to rate the system based on their experience and assessment. The responses were then collected and compiled by the researchers for data analysis.

Data Analysis Procedures

This section will describe how the collected data is going to be processed, analyzed and interpreted to support both system's automated recommendations and evaluation of effectiveness. The process combines of the structure data transformation, the system performance, analytics modeling and the respondent

based evaluation that will ensure the assessment of the An Skill-Extraction and Workload-Based Resource Recommendation System for WEA.

Data Processing and Preparation

The system will start by collecting the documents of the employee like certificate and CV, and using Optical Character Recognition or OCR to convert them into a system readable text and the data that is extracted in employee credential will undergoes preprocessing that includes the cleaning of unnecessary characters, normalizing of terms and removing the duplicates or informations that are irrelevant to the system. After that, the Natural Language Processing or NLP techniques are applied to the extracted data such as skills, certifications and experience levels of the employee, which are then stored in a database for an efficient analysis.

Skill Analysis and Structuring

The extracted data will be analyzed using Natural Language Processing or NLP techniques such as keyword matching, tokenization and synonym mapping to convert the different variations of skills into a consistent terms. The system will then organize the informations that are already processed into structured datasets, that includes the skill profiles, project requirement and experience classifications of the employees. This structured representation are enables accurate and efficient comparison between the employee qualifications and the specific needs of the project.

Matching and Recommendation Analysis

The resource manager will accept the project and the recommendation will start the system to perform matching analysis by comparing the employee profiles with the project requirements using mathematical models. It will determine the skill overlap, computes matching scores and ranks the employees based on their suitability to the project. Filtering mechanism will remove the candidates with no relevant skills to the project, while the allocation analysis calculates workload percentages based on the assigned and available working hours of the employee. These processes will ensure that only the employees that are qualified will be recommended and that the resources are distributed efficiently to the project.

Performance and Accuracy Analysis

To evaluate the effectiveness of the system, a quantitative analysis will be conducted using accuracy measure for both Optical Character Recognition or OCR and Natural Language Processing or NLP components. The OCR accuracy is determined by comparing the extracted text with the original document content of the employee, while the NLP accuracy will be evaluated by comparing the extracted skills with the manually identified data. The recommendation accuracy will be also assessed by verifying whether the suggested employees are meets the project requirements of the project. Error analysis is also performed to identify the causes of incorrect outputs and to distinguish between the system limitations and the data-related issues that may affect the overall performance of the system.

System-Based Data Analysis

The system based analysis will be focus on the overall performance of the automated components including the OCR, NLP and the recommendation engine. The accuracy of the system processes the data, the generates matching the score and the produces ranked recommendation will be analyze by the researchers. In evaluating the consistency and reliability, the quantitative metrics such as accuracy percentage, matching scores and allocation computation will be used. The result will be organize and will be process using the tools such as Python and Supabase to ensure the systematic tracking and validation of the output.

Respondent-Based Evaluation Analysis

The respondent based evaluation will be focusing on the assessing the system from the perspective of the selected evaluations using the ISO/IES 25010 based likert-scale questionnaire. By computing the mean scores for functional suitability, reliability and usability, the responses will be analyze quantitatively. With these scores will provide the clear measurement of the system quality based on the user experience.

Considering a feedback and observations from the respondents during system interaction, the qualitative interpretation is being incorporated. This will help to identify the usability concern and what areas needed to improve. The researchers will obtain balance evaluation both technical performance and user satisfaction by combining the quantitative ratings and the qualitative insights.

Ethical Considerations

Data Privacy and Confidentiality

The researchers ensures that all employee data that are extracted through OCR and processed through the NLP component is stored securely at all times. Personal information are encrypted both in the database and during transmission to prevent unauthorized access or data breaches. Access controls are also implemented so that only the authorized users can view or manage the sensitive information, that ensure the privacy of the employees whose documents are being process by the system.

Fairness and Bias Reduction

The recommendation algorithm of the system are design to provide fair and unbiased suggestions to the Resource Manager. Skill matching are based on actual qualifications and technical competencies of the employees, not on personal characteristics or other non-relevant factors. By doing this approach, equal opportunities are provided for all the employees so that the system will not favor or overlook anyone, which addresses the issue of underutilization that were identify in this study.

Transparency

The system is designed to provide clear explanations for every recommendations that it generates so the users can understand how the decisions are being made. The Resource Manager can see the basis of each

recommendation including the matched skills and the experience level of the employees. A documentation is also provided for all system functions to make sure everything is clear and understandable to the users.

Security

To protect the system from any unauthorized access, user authentication is implemented to ensure that only the authorized persons can use and access the system. Session management is also applied to prevent unauthorized use especially when the user forgot to log out. Input validation is also added to protect the system against any malicious attacks that can harm the data inside.

Academic Integrity

The researchers made sure that all the sources and references that are used in this study is properly cited and acknowledged following the APA citation format. The research are also follows the established development and documentation standards to maintains the quality and standard of the study. The testing results are reported accurately and without any manipulation to ensure the honesty and integrity of the findings of the study.

REFERENCE LIST

- Alberto, L., Lebis, A., Karami, A., Vermeulen, M., Fleury, A., & Van De Wiele, C. (2025). A recommender system of postgraduate courses based on soft skills: A psychometric-inspired approach. *International Journal of Artificial Intelligence in Education*. <https://doi.org/10.1007/s40593-025-00463-z>
- Alsaif, S. A., Hidri, M. S., Ferjani, I., Eleraky, H. A., & Hidri, A. (2022). NLP-based bidirectional recommendation system: Towards recommending jobs to job seekers and resumes to recruiters. *Big Data and Cognitive Computing*, 6(4), 147. <https://doi.org/10.3390/bdcc6040147>
- Andrade, C. (2021). The inconvenient truth about convenience and purposive samples. *Indian Journal of Psychological Medicine*, 43(1), 86–88. <https://doi.org/10.1177/0253717620977000>
- Bai, B., Dai, H., Zhang, D., Zhang, F., & Hu, H. (2022). The impacts of algorithmic work assignment on fairness perceptions and productivity: Evidence from field experiments. *Manufacturing & Service Operations Management*, 24(6), 3060–3078. <https://doi.org/10.1287/msom.2022.1120>
- Banerjee, A., Banik, P., & Wörndl, W. (2023). A review on individual and multistakeholder fairness in tourism recommender systems. *Frontiers in Big Data*, 6, Article 1168692. <https://doi.org/10.3389/fdata.2023.1168692>
- Barletta, V. S., Caivano, D., Colizzi, L., Dimauro, G., & Piattini, M. (2023). Clinical-chatbot AHP evaluation based on "quality in use" of ISO/IEC

25010. *International Journal of Medical Informatics*, 170, Article 104951.
<https://doi.org/10.1016/j.ijmedinf.2022.104951>

- Brill, K. (2025, July 9). What is Rapid Application Development (RAD)?
Built In. <https://builtin.com/articles/rapid-application-development>
- Cacatian, L. B., Esguerra, J. E. C., Estacaan, A. G., Marasigan, J. B., Sinosa, M. A. B., & Tatco, J. A. V. (2025). Automated innovation: Exploring the integration of artificial intelligence (AI) in human resource recruitment in Metro Manila. *ResearchGate*.
https://www.researchgate.net/publication/387787089_Automated_Innovation_Exploring_The_Integration_of_Artificial_Intelligence_AI_In_Human_Resource_Recruitment_in_Metro_Manila
- Carnovalini, F. (2025). Popularity bias in recommender systems. *Information*, 16(2), 151. <https://doi.org/10.3390/info16020151>
- De Campos, L. M., Fernandez-Luna, J. M., & Huete, J. F. (2024). An explainable content-based approach for recommender systems: A case study in journal recommendation for paper submission. *User Modeling and User-Adapted Interaction*. <https://doi.org/10.1007/s11257-024-09400-6>
- Gaur, B., Saluja, G. S., Sivakumar, H. B., & Singh, S. (2021). Semi-supervised deep learning based named entity recognition model to parse education section of resumes. *Neural Computing and Applications*, 33(11), 5705–5718. <https://doi.org/10.1007/s00521-020-05351-2>
- Gómez, E., Contreras, D., Boratto, L., & Salamó, M. (2025). Enhancing recommender systems with provider fairness through preference

distribution awareness. *International Journal of Information Management Data Insights*, 5(1), Article 100311.

<https://doi.org/10.1016/j.jjime.2024.100311>

- Gotz, G. (2023, December 29). *An approach to implement user-based recommendation systems with small-sized data sets*. ResearchGate. <https://www.researchgate.net/publication/376819602>
- ISO/IEC/IEEE. (2022). *ISO/IEC/IEEE 26514:2022 — Systems and software engineering: Design and development of information for users*. International Organization for Standardization. <https://www.iso.org/standard/79060.html>
- Janani, M., Laya, H., & Mathishree, B. (2025). OCR text recognition and recommendation using machine learning. In *2025 International Conference on Visual Analytics and Data Visualization (ICVADV)*. IEEE. https://www.researchgate.net/publication/391038785_OCR_Text_Recognition_and_Recommendation_Using_Machine_Learning
- Jin, D., Wang, L., Zhang, H., Zheng, Y., Ding, W., Xia, F., & Pan, S. (2023). A survey on fairness-aware recommender systems. *arXiv*. <https://doi.org/10.48550/arXiv.2306.00403>
- Kasri, M. A., Muharom, L. A., & Wardani, R. C. (2024, April 12). Evaluation of the new student candidates admission information system using ISO/IEC 25010 model. *OSF Preprints*. <https://doi.org/10.31219/osf.io/e8znx>

- Li, Y., Chen, H., Xu, S., Ge, Y., Tan, J., Liu, S., & Zhang, Y. (2022). Fairness in recommendation: A survey. *arXiv*. <https://doi.org/10.48550/arXiv.2205.13619>
- Lima, S. A., & Rahman, M. M. (2024). Algorithmic fairness in HRM: Balancing AI-driven decision making with inclusive workforce practices. *Journal of Information Systems Engineering and Management*, 9(4s). <https://doi.org/10.52783/jisem.v9i4s.13251>
- Malashin, I., Masich, I., Tynchenko, V., Gantimurov, A., Nelyub, V., & Borodulin, A. (2024). Image text extraction and natural language processing of unstructured data from medical reports. *Machine Learning and Knowledge Extraction*, 6(2), 1361–1377. <https://doi.org/10.3390/make6020064>
- Meyliana, A., Rapiyanta, P. T., & Andriani, A. (2024). Application of the Rapid Application Development (RAD) method for web-based financial management and wood inventory using CodeIgniter. *ARRUS Journal of Engineering and Technology*, 4(1), 81–89. <https://doi.org/10.35877/jetech2722>
- Najam, R., & Faizullah, S. (2023). Analysis of recent deep learning techniques for Arabic handwritten-text OCR and post-OCR correction. *Applied Sciences*, 13(13), 7568. <https://doi.org/10.3390/app13137568>
- Novita, D., Rapiyanta, P. T., & Andriani, A. (2023). Implementation of Rapid Application Development in Android-based school information

system. Raden Intan State Islamic University.
<https://ejournal.radenintan.ac.id>

- Nugraha, U., & Rachman, M. A. N. (2024, December 29). Quality evaluation study. In *Proceedings of the Widyatama International Conference on Engineering 2024 (WICOENG 2024)*. Atlantis Press.
<https://www.atlantis-press.com/proceedings/wicoeng-24/126007237>
- Nuzula, M. I. F., & Rochimah, S. (2023). Evaluation of service quality in human resource information systems using the ISO/IEC 25010. In *2023 International Seminar on Application for Technology of Information and Communication (iSemantic)* (pp. 219–224). IEEE.
<https://doi.org/10.1109/iSemantic59612.2023.10295365>
- Protasiewicz, J., Podwysocki, E., Ostrowska, S., & Tomczyńska, A. (2021). Open access to data on higher education and science: A case study of the RAD-on platform in Poland. In *Proceedings of the European University Information Systems Conference 2021* (Vol. 78, pp. 9–21). EasyChair. <https://easychair.org/publications/paper/6pSZ>
- Rampisela, T. V., Maistro, M., Ruotsalo, T., & Lioma, C. (2023). Evaluation measures of individual item fairness for recommender systems: A critical study. *arXiv*. <https://doi.org/10.48550/arXiv.2311.01013>
- Roy, D., & Dutta, M. (2022). A systematic review and research perspective on recommender systems. *Journal of Big Data*, 9(1), 59.
<https://doi.org/10.1186/s40537-022-00592-5>

- Deshmukh, A., & Raut, A. (2024). Applying BERT-based NLP for automated resume screening and candidate ranking. *Annals of Data Science*. <https://doi.org/10.1007/s40745-024-00524-5>
- Saatçı, M., Kaya, R., & Ünlü, R. (2024). Resume screening with natural language processing (NLP). *Alphanumeric Journal*, 12(2), 121–140. <https://doi.org/10.17093/alphanumeric.1536577>
- Wang, L., Mistry, S., Hasan, A. A., Hassan, A. O., Islam, Y., & Osei, F. A. J. (2023). Implementation of a collaborative recommendation system based on multi-clustering. *Mathematics*, 11(6), 1346. <https://doi.org/10.3390/math11061346>
- Wang, Y. (2025). Improving user-oriented fairness in recommendation systems. *Journal of Systems and Software*, 223, Article 112387. <https://doi.org/10.1016/j.jss.2025.112387>
- Varshith Reddy, K. S., Kiran, N. J., Likhith Gowda, T. R., Amanullah, M., & Harini, S. (2025). Resume analyzer and job recommendation system. *Iconic Research and Engineering Journals*, 8(9), 282–287. <https://www.irejournals.com/paper-details/1707439>
- Zhou, X., Zhang, L., Zhang, H., Zhang, Y., Zhang, X., Zhang, J., & Shen, Z. (2024). Advancing sustainability via recommender systems: A survey. *arXiv*. <https://doi.org/10.48550/arXiv.2411.07658>
- Mueller, V., & Schaaff, K. (2025). AI-based recommender system for employee-project matching of IT specialists. In T. Schlippe, E. C. K. Cheng, & T. Wang (Eds.), *Artificial intelligence in education technologies:*

New development and innovative practices (pp. 296–311). Springer.
https://doi.org/10.1007/978-981-97-9255-9_20

- Tang, F., Zhu, R., Yao, F., Wang, J., Luo, L., & Li, B. (2025). Explainable person–job recommendations: Challenges, approaches, and comparative analysis. *Frontiers in Artificial Intelligence*, 8, Article 1660548.
<https://doi.org/10.3389/frai.2025.1660548>
- Boukhari, S., Tlili, M., & Abdellatif, T. (2024). Skill matching at scale: Freelancer-project alignment for efficient multilingual candidate retrieval. *arXiv*. <https://arxiv.org/abs/2409.12097>
- Kashif, M., & Kumar, K. R. P. (2024). NLP-powered resume matching for recruitment. *International Journal of Advanced Research in Computer and Communication Engineering*, 13(9), 106–113.
<https://www.ijfmr.com/papers/2024/6/31742.pdf>
- Kishore, C. R., & Sreerala, T. (2025). AI-powered resume analyzer and job matching system. *International Journal of Research in Intelligent Systems*, 10(11), 382–387. <https://doi.org/10.51584/IJRIAS>
- Naveed, Z., Nisar, B., Saifullah, D. M., & Iqbal Baig, J. (2024). Resume ranking using natural language processing. *Journal of Computers and Intelligent Systems*, 2(1), 61–70.
<https://dergipark.org.tr/en/download/article-file/4159614>
- Decorte, J. J., Van Haute, J., Demeester, T., & Develder, C. (2024). SkillMatch: Evaluating self-supervised learning of skill relatedness. *arXiv*.
<https://arxiv.org/abs/2410.05006>

- Clavié, B., & Soulié, G. (2023). Large language models as batteries-included zero-shot ESCO skills matchers. *arXiv*. <https://arxiv.org/abs/2307.03539>
- Magron, A., Dai, A., Zhang, M., Montariol, S., & Bosselut, A. (2024). JobScape: A framework for generating synthetic job postings to enhance skill matching. In E. Hruschka, T. Lake, N. Otani, & T. Mitchell (Eds.), *Proceedings of the first workshop on natural language processing for human resources (NLP4HR 2024)* (pp. 43–58). Association for Computational Linguistics. <https://aclanthology.org/2024.nlp4hr-1.4/>
- Gavrilescu, M., Leon, F., & Minea, A. A. (2025). Techniques for transversal skill classification and relevant keyword extraction from job advertisements. *Information*, 16(2), 167. <https://doi.org/10.3390/info16020167>
- Helli, S. S., Tanberk, S., & Cavsak, S. N. (2023). Resume information extraction via post-OCR text processing. In *2023 IEEE 3rd Mysore Sub Section International Conference (MysuruCon)* (pp. 1–6). IEEE. <https://doi.org/10.1109/MysuruCon59703.2023.10296715>
- Thangam, S., Kumaran, U., Biswas, D., Sneha, B., Nadipalli, S., & Raja, S. (2023). Text post-processing on optical character recognition output using natural language processing methods. In *2023 IEEE 3rd Mysore Sub Section International Conference (MysuruCon)* (pp. 1–6). IEEE. <https://doi.org/10.1109/MysuruCon59703.2023.10296715>

- Parlika, R., Afifudin, M., Pradana, I. A., Wiratama, Y. D. W., & Holis, M. N. (2023). Literature study on the efficiency of the Rapid Application Development model in software development. *POSITIF: Jurnal Sistem dan Teknologi Informasi*, 8(2), 64–73.
<https://doi.org/10.31961/positif.v8i2.1329>
- Riadi, I., Yudhana, A., & Elvina, A. (2024). Analysis impact of Rapid Application Development method on development cycle and user satisfaction: A case study on web-based registration service. *Scientific Journal of Informatics*, 11(1).
<https://journal.unnes.ac.id/nju/sji/article/view/49590>
- Stankovic, J. (2024). Rapid Application Development in modern software engineering. *ResearchGate*.
<https://www.researchgate.net/publication/379548372>
- Imran, M., & Bhutto, A. (2023). RAD-based web application development for small and medium enterprises. *International Journal of Scientific Research in Computer Science Engineering and Information Technology*, 9(3). <https://ijsrcseit.com/paper/CSEIT2390345.pdf>
- Rojas, H., Renteria, R., Duran, V. M., Gutiérrez, Y. T., Cabrera, M. J. I., & Aminuddin, A. (2025). Mapping the evolution and future directions of ISO/IEC 25010: A bibliometric and thematic analysis. *Engineering, Technology & Applied Science Research*, 15(5), 27530–27541.
<https://doi.org/10.48084/etasr.11772>

- Ilyas, M., & Ahmad, W. (2023). Evaluation of web-based information system quality using ISO/IEC 25010. *International Journal of Information System and Technology*, 7(2), 234–241.
<https://ijistech.org/ijistech/index.php/ijistech/article/view/234>
- Susanto, A., & Andriana, D. (2023). Quality measurement of academic information system based on ISO/IEC 25010. *Journal of Informatics and Applied Mathematics*, 6(1).
<https://journal.stmikjayakarta.ac.id/index.php/jisamar/article/view/819>
- Zhao, Y., Wang, Y., Liu, Y., Cheng, X., Aggarwal, C. C., & Derr, T. (2025). Fairness and diversity in recommender systems: A survey. *ACM Transactions on Intelligent Systems and Technology*, 16(1), Article 2, 1–28. <https://doi.org/10.1145/3664928>